

Customer Shopping Behavior Analysis

3,900 purchases — 18 features — goal: uncover spending patterns, segments, product & subscription insights.



Dataset Summary

Rows / Columns

3,900 rows • 18 columns

Key Features

Demographics, Purchase details, Behavior, Shipping, Discounts

Missing Data

37 missing values in Review Rating



Data Preparation (Python)

01

Load & Inspect

pandas: df.info(), .describe()

02

Missing Values

Impute Review Rating by category median

03

Standardize & Engineer

snake_case; age_group; purchase_frequency_days

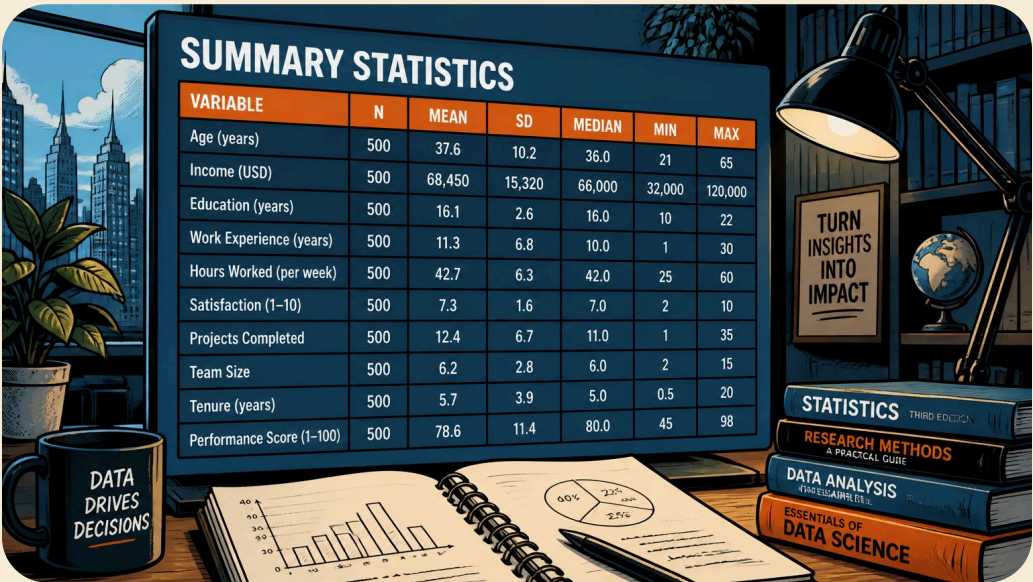
04

DB Integration

Loaded cleaned df into PostgreSQL



Exploratory Stats Snapshot



Count: 3,900 · Mean age: 44.07 · Avg purchase: \$59.76 · Avg rating: 3.75

Top gender: Male (2,652) · Top item: Blouse (171)

SQL Findings — Revenue & Segments



Revenue by Gender

Male: 157,890 • Female: 75,191



Customer Segments

Loyal: 3,116 • Returning: 701 • New: 83



Subscribers

Yes: 1,053 (avg \$59.49, revenue 62,645) •
No: 2,847 (avg \$59.87, revenue 170,436)

Product & Discount Insights



Discount-Dependent

Hat 50.00% • Sneakers 49.66% • Coat
49.07% • Sweater 48.17% • Pants 47.37%

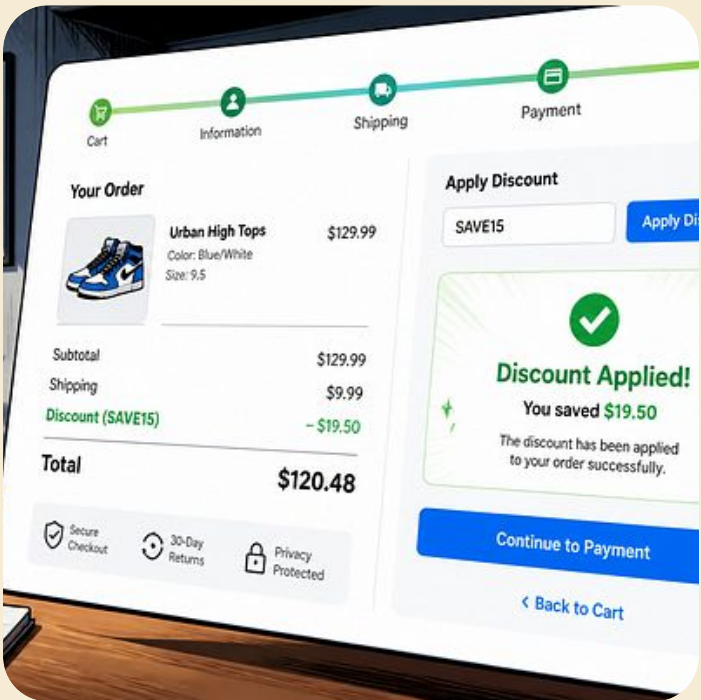


Top Rated Items

Gloves 3.86 • Sandals 3.84 • Boots 3.82 •
Hat 3.80 • Skirt 3.78

High-Spending Discount Users

839 rows flagged where customers used discounts yet spent above average.



Age & Revenue Dynamics

62K

Young Adult

59K

Middle-aged

56K

Adult

55K

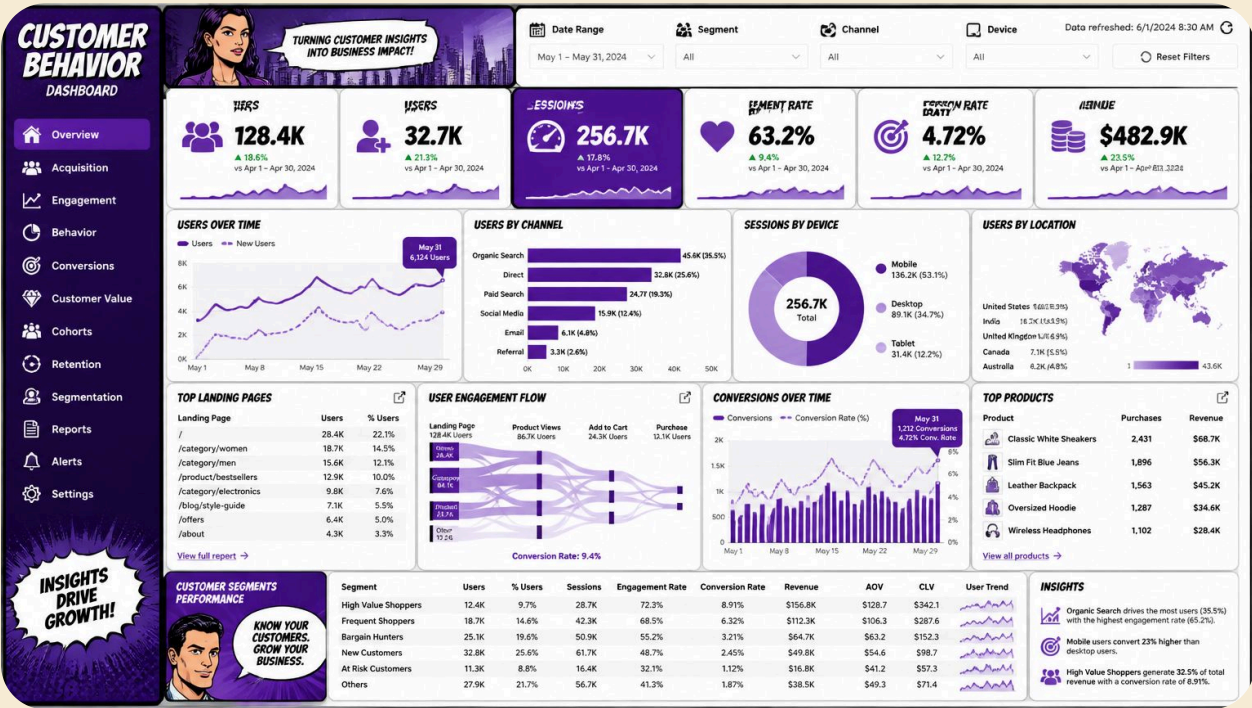
Senior

Revenue by age group — target high-revenue cohorts.



Power BI Dashboard

Interactive visuals: key metrics, category revenue, sales, age & subscription breakdowns.



Business Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits

Loyalty Programs

Reward repeat buyers to grow Loyal segment

Review Discounts

Balance promotional lift with margins

Product Positioning

Highlight top-rated & best-selling items in campaigns

Targeted Marketing

Focus on high-revenue age groups & express-shipping users

📋 Next steps: test subscription offers, A/B discount strategies, and personalize campaigns by segment.